Does Causal Structure Leave a Signature in Sparse Autoencoder Representations?

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Abstract

Sparse autoencoders (SAEs) are widely used to extract interpretable features from neural network activations, but it is unclear whether their learned representations are shaped only by observable data statistics or also by the underlying generative structure that produced those statistics. We investigate this question using two synthetic datasets with matched observable statistics (correlation, joint probabilities, sparsity) but different causal structures: a direct-dependency world (A $\rightarrow$ B) and a common-cause world (C $\rightarrow$ {A, B}), where the shared cause C is never observed by the model. We train SAEs on each world across multiple random seeds and compare the resulting representations using four metrics: raw feature-recovery correlation, partial correlation controlling for the A–B relationship, best-latent-per-seed comparison, and aggregate representation geometry. Across all four metrics, the two worlds produce statistically indistinguishable representations, with differences consistently within normal seed-to-seed variance. This result is consistent with the hypothesis that, under these controlled conditions, SAE representations are driven by observable statistics rather than by the underlying causal mechanism – a negative result that clarifies a boundary condition for representation learning research, while leaving open whether larger or nonlinear architectures would behave differently.

1 Introduction

Sparse autoencoders (SAEs) are widely used for interpretability research, extracting features from neural network activations [Bricken et al., 2023, Cunningham et al., 2023]. They are trained on observable data, but real-world data reflects both statistical patterns and the causal mechanisms that generate them. This raises a question: do SAEs pick up on causal structure, leaving a consistent signature behind, or do they only reflect surface-level statistics? Distinguishing between these possibilities matters for understanding how much SAE-derived features can be trusted as capturing genuine underlying mechanisms, rather than merely correlational patterns. Our work investigates this by constructing two synthetic "worlds" with matched observable statistics but different underlying causal structures, and testing whether SAEs trained on each produce different representations. We find that, under these controlled conditions, SAE representations for both causal structures are statistically indistinguishable across four independent metrics.

2 Experimental Setup

Data generation. World A features a direct dependency between A and B, where B's value depends on A. World B has no direct dependency between A and B; instead, both are generated from a shared confounder C, which is never observed by the model. Four additional independent distractor features were included in both worlds, since real-world data typically contains noise unrelated to the variables

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of interest; if causal structure genuinely leaves a detectable signature, it should remain identifiable even in the presence of such noise.

Model architecture. The SAE consists of a linear encoder mapping the 6-dimensional input to a 20-dimensional latent space, followed by a ReLU nonlinearity, and a linear decoder mapping back to 6 dimensions. The latent space is intentionally overcomplete (20 > 6), since the goal is not dimensional compression but a sparse, expressive basis in which only a small subset of latents are active for any given input [Elhage et al., 2022]. ReLU enforces this structurally by zeroing out negative activations.

Loss function. The loss function consists of two terms. The first is Mean Squared Error (MSE), the average of squared differences between the original and reconstructed values, measuring reconstruction quality. The second is $\lambda \cdot ||z||_1$, where $||z||_1$ (the L1 norm) is the sum of the absolute values of all latent activations, and λ is a coefficient controlling the tradeoff between the two terms. A higher λ increases pressure toward sparsity, potentially at the cost of reconstruction accuracy; a lower λ reduces that pressure, and if set too low, undermines the purpose of a sparse autoencoder specifically.

The full objective is:

$$L = L_{recon} + \lambda \cdot ||z||_1$$

Training. For each world, five SAEs were trained independently for 200 epochs using the Adam optimizer with a learning rate of 1×10^{-3}. Each of the five runs used a distinct random seed controlling model weight initialization, while the underlying dataset was held fixed (1000 samples, data-generation seed = 42) across all five runs within a world. This design isolates variation due to training dynamics from variation due to the data itself. The sparsity weight was set to $\lambda = 0.03$ for both worlds, selected via a sweep described in Section 3.

3 Statistical Validation

The two worlds' observable statistics were matched via a parameter sweep on World B's confounder-conditional probabilities, targeting World A's correlation between A and B. Table 1 shows the resulting statistics for both worlds (1000 samples, seed=42).

The match was validated across 5 additional random seeds, confirming it was not specific to a single lucky draw: the mean absolute difference in corr(A,B) between the two worlds across seeds was approximately 0.03, small relative to the observed correlation values themselves (≈0.75–0.80).

Table 1: Observable statistics for World A and World B (1000 samples, seed=42).

World	P(A)	P(B)	P(A,B)	corr(A,B)
A	0.304	0.333	0.270	0.779
B	0.324	0.317	0.270	0.768

4 Results

4.1 Feature Recovery

Correlating each latent's activation with ground-truth A and B revealed that no latent tracked either variable in isolation; instead, the strongest latents (e.g., latent 12: corr(A) = 0.82, corr(B) = 0.88 in World A) tracked both simultaneously. This pattern was nearly identical in World B (latent 12: corr(A) = 0.81, corr(B) = 0.89), which is expected given that A and B are correlated with each other in both worlds by construction (Section 3). Several latents (typically 3 of 20) were "dead" – never activating across the dataset – and were excluded from correlation analysis.

4.2 Partial Correlation

Since raw correlation cannot distinguish a latent genuinely tracking A from one merely reflecting the shared A–B correlation, partial correlation was used to control for each variable when assessing the other. An initial analysis averaged partial correlations across seeds by raw latent index, but this was identified as methodologically invalid: latent index is not consistent across independently-initialized models, so averaging by index mixes unrelated latents. This was corrected by instead comparing, for each seed, the single best-matching latent for A and for B.

Figure 1 shows this comparison across all 5 seeds. World A and World B track each other closely in both panels, with differences (mean absolute difference ≈0.03–0.06) consistent with ordinary seed-to-seed variance rather than a systematic gap between worlds.

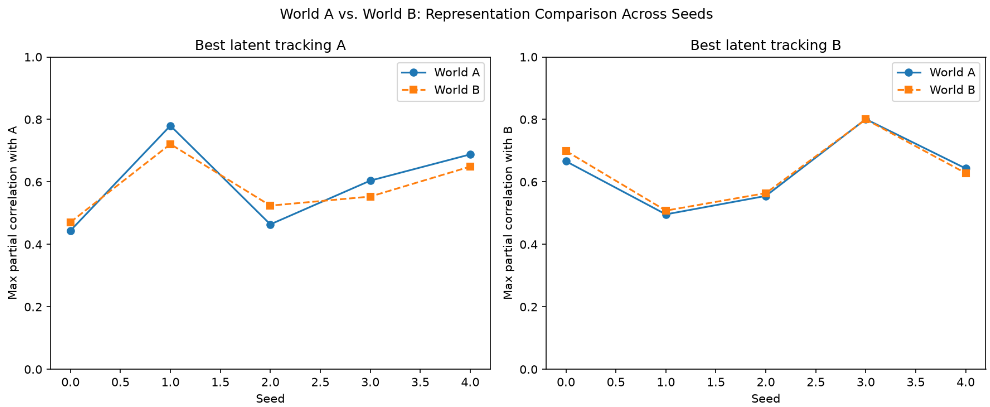

Figure 1: Best-latent-per-seed partial correlation with A (left) and B (right), World A vs. World B, across 5 seeds. The two worlds track each other closely, with no systematic gap.

4.3 Representation Geometry

As a fourth, structurally distinct check, the mean absolute off-diagonal correlation between latents (a summary of overall inter-latent entanglement) was compared across worlds. World A: [0.251, 0.279, 0.240, 0.280, 0.291] across 5 seeds; World B: [0.246, 0.275, 0.234, 0.283, 0.291]. Differences ranged from 0.000 to 0.006 – tighter than any other metric tested.

4.4 Sparsity–Reconstruction Tradeoff

Figure 2 shows the tradeoff used to select $\lambda = 0.03$, sweeping over [0.01, 0.03, 0.05, 0.07, 0.1] on World A. Higher λ improved sparsity at the cost of reconstruction accuracy, as expected from the loss formulation in Section 2.

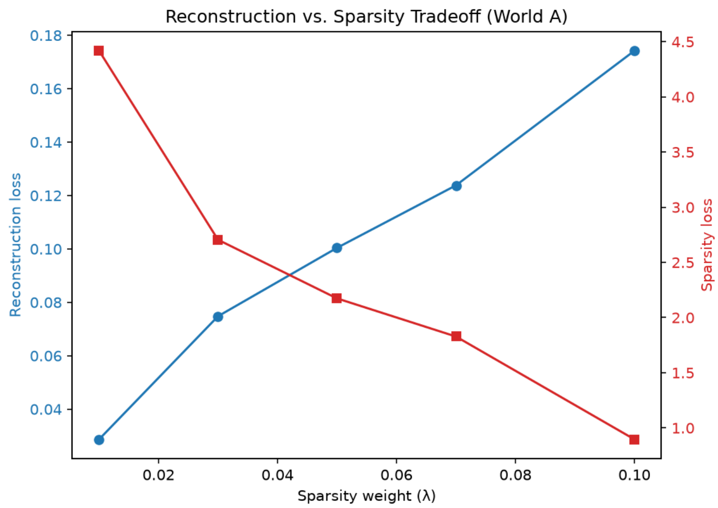

Figure 2: Reconstruction loss and sparsity loss as a function of λ (World A). $\lambda = 0.03$ was selected as a reasonable middle ground.

5 Discussion

Across both worlds, results were statistically indistinguishable across all four metrics. This indicates that, under these conditions, the SAE's representation did not reflect a meaningful signature of causal structure – regardless of whether the underlying relationship was a direct dependency or a confound-induced correlation, the learned representation looked the same. Since the two worlds' observable statistics were deliberately matched, causal mechanism was the only remaining variable that could have differed between them – yet even this left no detectable trace. In this setup, the SAE's representation did not encode information sufficient to distinguish a direct dependency from a shared confound, once their observable statistics were matched.

This matters practically for interpretability research: if an SAE's learned features are used as evidence for a particular causal mechanism, this result suggests such an inference may not be warranted. A given representation does not reliably indicate what kind of causal mechanism produced it – the same representation can arise from either a direct dependency or a shared confound. Under these conditions, representation similarity alone is not decisive evidence for any one particular mechanism.

6 Limitations

This work has several limitations that constrain how broadly the result should be interpreted.

Synthetic, low-dimensional data. The experiments used synthetic binary features (6 total dimensions), not real neural network activations. Real-world representations are typically continuous, high-dimensional, and involve far more complex feature interactions than the two-variable causal structures tested here.

Small, linear architecture. The SAE used a single linear encoder and decoder. Many SAEs used in practice for interpretability research involve larger, and sometimes nonlinear, architectures; results may differ for such models.

Single matching strength tested. Observable statistics were matched at one particular correlation strength (corr(A,B) ≈ 0.77). Whether the result holds at different matching strengths, or under weaker/stronger statistical constraints, was not tested.

Binary features only. All variables were binary (Bernoulli-distributed). Continuous features, as found in real activations, were not tested and may reveal different representational behavior.

Correlation-based metrics. The four metrics used (feature recovery, partial correlation, best-latent comparison, geometry) are primarily linear/correlation-based. Nonlinear or higher-order representational differences between the two worlds, if present, may not have been detected by these metrics.

Code availability. All code, data generation scripts, and analysis notebooks are available at `https://github.com/Malhar379/sae-causal-signatures`.

7 Future Work

Natural extensions of this work include testing whether the result holds with nonlinear SAE architectures, real neural network activations rather than synthetic data, continuous rather than binary features, and a range of statistical matching strengths rather than a single fixed correlation target. Testing more extreme causal structures (e.g., higher-order confounding, multiple interacting causal chains) may also reveal conditions under which representational differences do emerge.

References

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